

Centers of Mass & Volumes of Revolution

MATH 146 – Calculus II

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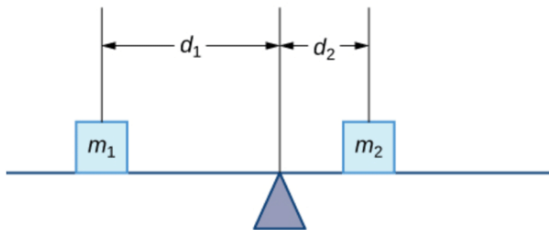
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Archimedes' Law of the Lever

Theorem (Archimedes' Law of the Lever)

Two magnitudes balance at distances reciprocally proportional to their magnitudes.



In other words, $\frac{d_1}{d_2} = \frac{m_2}{m_1} \iff d_1 m_1 = d_2 m_2$.

Centers of Gravity for Some Elementary 2D Shapes

Archimedes showed that:

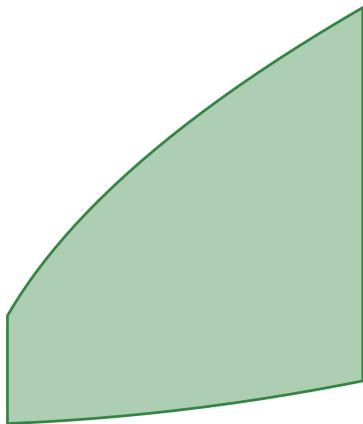
The center of gravity of a parallelogram is the point of intersection of the diagonals.

The center of gravity of a triangle lies on the straight line between a vertex and the center of the opposite side.

The center of gravity of a parabolic segment lies at a point G on the diameter of the parabola which is $\frac{3}{5}$ of the way from the vertex to the base.

Center of Gravity

Given any region bounded by $f(x)$ and $g(x)$, can we find the coordinates $(\bar{x}, \bar{y})$ of its center of gravity?



The Moment Equation

Theorem (The Moment Equation)

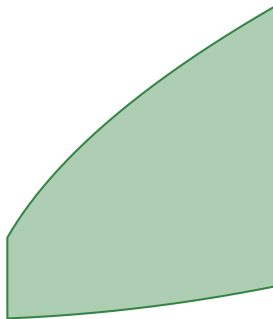
If masses $m_1, m_2, \dots, m_n$ are suspended on a balance beam from points $x_1, x_2, \dots, x_n$, then the system will be in balance when suspended from the point $\bar{x}$ such that:

$$\sum_{i=1}^n m_i(\bar{x} - x_i) = 0,$$

where:

$$\bar{x} = \frac{\sum_{i=1}^n m_i x_i}{\sum_{i=1}^n m_i}$$

Center of Gravity

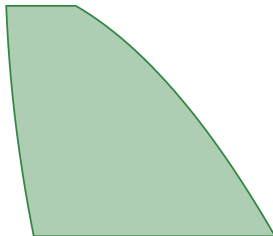


$$\bar{x} = \frac{\sum_{i=1}^n m_i x_i}{\sum_{i=1}^n m_i} \quad \Rightarrow \quad \bar{x}_P = \frac{\sum_{i=1}^n x_i^* (f(x_i^*) - g(x_i^*)) \Delta x_i}{\sum_{i=1}^n (f(x_i^*) - g(x_i^*)) \Delta x_i}$$

If we send the width of the intervals to 0, we can rewrite these as integrals.

$$\bar{x}_P = \lim_{\Delta x \rightarrow 0} \frac{\sum_{i=1}^n x_i^* (f(x_i^*) - g(x_i^*)) \Delta x_i}{\sum_{i=1}^n (f(x_i^*) - g(x_i^*)) \Delta x_i} = \frac{\int_a^b x (f(x) - g(x)) \, dx}{\int_a^b f(x) - g(x) \, dx}$$

Center of Gravity



$$\bar{y} = \frac{\sum_{i=1}^n m_i y_i}{\sum_{i=1}^n m_i} \implies \bar{y}_P = \frac{\sum_{i=1}^n \left(\frac{f(x_i^*) + g(x_i^*)}{2} \right) (f(x_i^*) - g(x_i^*)) \Delta x_i}{\sum_{i=1}^n (f(x_i^*) - g(x_i^*)) \Delta x_i}$$

Center of Gravity

If we send the width of the intervals to 0, we can rewrite these as intervals.

$$\begin{aligned}\bar{y}_P &= \lim_{\Delta x \rightarrow 0} \frac{\sum_{i=1}^n \left(\frac{f(x_i^*) + g(x_i^*)}{2} \right) (f(x_i^*) - g(x_i^*)) \Delta x_i}{\sum_{i=1}^n (f(x_i^*) - g(x_i^*)) \Delta x_i} \\ &= \frac{1}{2} \left(\frac{\int_a^b f(x)^2 - g(x)^2 dx}{\int_a^b f(x) - g(x) dx} \right)\end{aligned}$$

Center of Gravity

Theorem

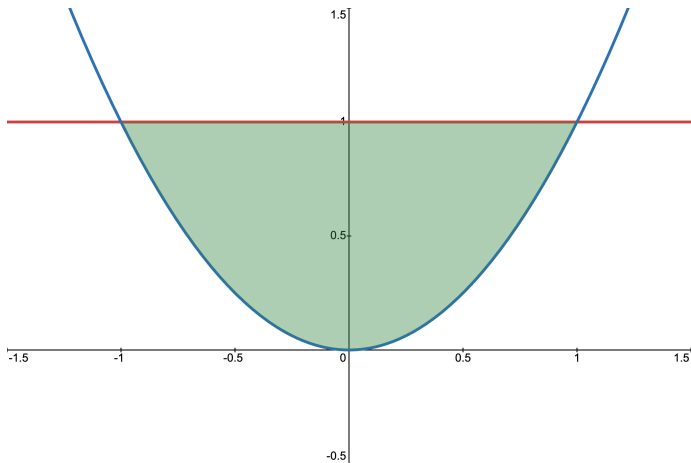
If a region R is bounded by the graphs of $f(x) > g(x)$ on the interval $[a, b]$, then the center of gravity of R can be found at the point with coordinates $(\bar{x}, \bar{y})$, where:

$$\bar{x} = \frac{\int_a^b x (f(x) - g(x)) \, dx}{\int_a^b f(x) - g(x) \, dx}$$

$$\bar{y} = \frac{\int_a^b f(x)^2 - g(x)^2 \, dx}{2 \int_a^b f(x) - g(x) \, dx}$$

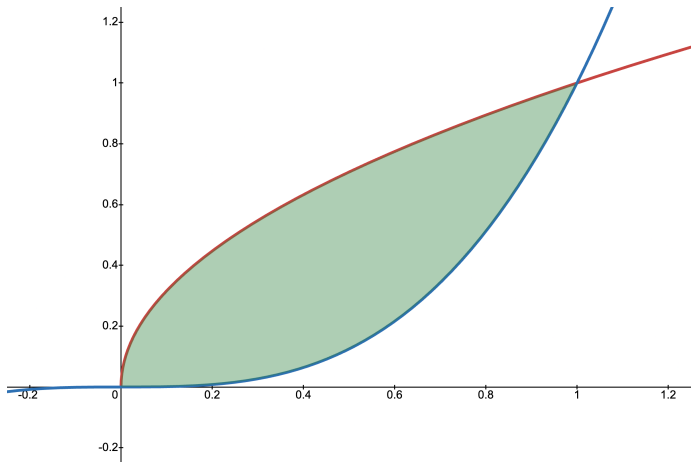
Example: Center of Gravity

Example: Find the center of gravity of the region bounded by the curves $y = 1$ and $y = x^2$.



Example: Center of Gravity

Example: Find the center of gravity of the region bounded by the curves $y = \sqrt{x}$ and $y = x^3$.

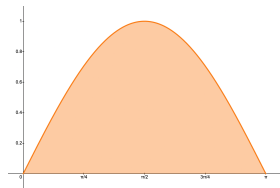


Volume of Revolution

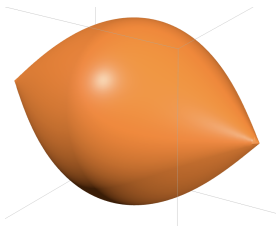
Volume of Revolution

Given a curve $f(x) \geq 0$ over the interval $[a, b]$, a **volume of revolution** is the 3D figure obtained by revolving the region between $f(x)$ and the x -axis about an axis.

Example: $f(x) = \sin(x)$ on the interval $x \in [0, \pi]$.



(a) xy -plane



(b) x -axis



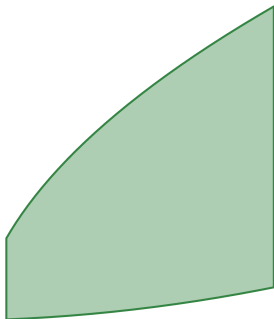
(c) y -axis

How do we calculate the volume of these?

The Pappus-Guldin Theorem

The Pappus-Guldin Theorem

Let R be a plane region that lies entirely on one side of a line l in the plane. If R is rotated about l , then the volume of the resulting solid is the product of the area A of R and the distance traveled by the centroid of R .



The Disk/Washer Method

The Disk Method (around the x -axis)

Suppose $f(x) \geq g(x) \geq 0$ over an interval $[a, b]$. Then the volume of revolution obtained by rotating the region contained by $f(x)$ and $g(x)$ over the interval $[a, b]$ around the x -axis is:

$$V = \pi \int_a^b f(x)^2 - g(x)^2 dx$$

$$\begin{aligned} V &= (2\pi\bar{y}) \times A_R \\ &= 2\pi \left(\frac{\int_a^b f(x)^2 - g(x)^2 dx}{2 \int_a^b f(x) - g(x) dx} \right) \left(\int_a^b f(x) - g(x) dx \right) \end{aligned}$$

Example: The Disk/Washer Method

Example: Find the volume of revolution obtained by revolving the region beneath the curve $y = x^2$ around the x -axis with $0 \leq x \leq 1$.

Example: Find the volume of revolution obtained by revolving the region beneath the curves $y = x^2$ and $y = 1$ around the x -axis with $0 \leq x \leq 1$.

The Shell Method

The Shell Method (around the y -axis)

Suppose $f(x) \geq g(x) \geq 0$ over an interval $[a, b]$. Then the volume of revolution obtained by rotating the region contained by $f(x)$ and $g(x)$ over the interval $[a, b]$ around the y -axis is:

$$V = 2\pi \int_a^b x (f(x) - g(x)) \, dx$$

$$\begin{aligned} V &= (2\pi \bar{x}) \times A_R \\ &= 2\pi \left(\frac{\int_a^b x (f(x) - g(x)) \, dx}{\int_a^b f(x) - g(x) \, dx} \right) \left(\int_a^b f(x) - g(x) \, dx \right) \end{aligned}$$

Example: The Shell Method

Example: Find the volume of revolution obtained by revolving the region beneath the curve $y = x^2$ around the x -axis with $0 \leq y \leq 1$.

Example: Find the volume of revolution obtained by revolving the region beneath the curves $y = x^2$ and $y = 1$ around the y -axis with $0 \leq x \leq 1$.